

TarunSQL Corp: Executive Dashboard Project

Technical Requirements & Problem Statements

1. Executive Summary

The executive leadership of TarunSQL Corp has commissioned a data-driven dashboard to support strategic decision-making regarding infrastructure expansion and human resource management. This document outlines the technical requirements for the SQL-based analysis.

2. Geographic Expansion Analysis

Business Context: Identification of high-density user regions to determine optimal locations for new office branches.

Problem Statement: Develop a query to identify the top three cities with the highest volume of registered users.

- Deliverables: Report columns must include City, State, and Total_Users.
- Technical Hint: Utilization of JOIN operations between 'users' and 'addresses' tables is expected.

3. Compensation & Equity Audit

Business Context: Analysis of salary distribution to ensure competitive compensation across demographics.

Problem Statement: Isolate the second-highest earner within each gender category to evaluate salary benchmarking.

- Deliverables: Report columns must include Name, Gender, Salary, and Rank.
- Technical Hint: Implementation of Window Functions (DENSE_RANK) partitioned by demographic groups.

4. User Interface Preference Analysis

Business Context: Assessment of user customization trends to guide UI/UX development.

Problem Statement: Extract and list users who have explicitly configured their interface settings to 'Dark Mode'.

- Deliverables: Target JSON configuration objects where 'theme' is set to 'dark'.
- Technical Hint: Usage of JSON extraction operators (->>) is required.

5. Internal Referral Network Mapping

Business Context: Visualization of the internal talent acquisition hierarchy. □

Problem Statement: Generate a hierarchical report displaying employees alongside their respective referrers. The dataset must strictly filter for employees who were acquired through the referral program.

Technical Hint: Execution of a Self-Join on the user directory.

6. Query Performance Optimization

Business Context: Technical audit of query efficiency to ensure scalability.

Problem Statement: Analyze the execution plan of the Geographic Expansion Analysis (Task 1) to verify index utilization (Primary Key vs. Full Table Scan).

